OAK INTERNATIONAL SCHOOL

PREMIUM MATHEMATICS ACTIVITY WORKBOOK — CLASS 4

Grade Level: Quatrième (4ème) / IB MYP 4 (Ages 13–14)

Academic Year: 2026–2027 | **Series Version:** OIS-WB-4EME-MYP4-V1.0

Programmes: IB Middle Years Programme (English) & Beninese National APC Curriculum (Français)

SECTION A — STUDENT INTRODUCTION / INTRODUCTION POUR L'ÉLÈVE

Entering the Golden Age of Geometry and Algebraic Proof

Bienvenue en classe de Quatrième / IB MYP 4 à Oak International School !

In Quatrième / MYP 4, your mathematical power reaches an extraordinary turning point:

- In **IB MYP 4**, you harness the universal power of **Pythagoras' Theorem** in 2D and 3D, expand your algebraic toolset with binomial expansions and linear equations, and connect right-triangle trigonometry (\sin, \cos, \tan) to navigation and engineering.
- In **Beninese 4ème (APC)**, you explore **Situation d'Apprentissage N°1 (Configurations du plan)** and **N°4 (Applications du plan)**. You master the converse of Pythagoras to prove right angles with flawless deductive rigor, unlock the geometric beauty of vectors (\vec{u}) and Chasles' relation, and uncover the centroid (G) and orthocentre (H) of triangles.

This workbook harmonises both traditions into a single, high-performance learning companion.

SECTION B — BILINGUAL MATHEMATICAL LANGUAGE / GLOSSAIRE BILINGUE

Pythagoras, Vectors & Literal Algebra

English Term	French Equivalent (*Terme Français*)	Mathematical Definition & Symbolic Example	Common Translation Pitfalls
Hypotenuse	*Hypoténuse*	The longest side of a right-angled triangle, opposite the 90^\circ angle.	Spelling check: *hypoténuse* in French with one 'h'; English has 'h' at start only.
Pythagoras' Theorem	*Théorème de Pythagore*	In a right triangle with hypotenuse c: a^2 + b^2 = c^2.	In French, strictly distinguish the direct theorem from its converse (*réciproque*).
Converse theorem	*Réciproque d'un théorème*	The reverse logical implication: If BC^2 = AB^2 + AC^2, then triangle ABC is right-angled at A.	Do not confuse *réciproque* (converse) with *contraposée* (contrapositive).
Cosine ratio	*Cosinus d'un angle aigu*	In a right-angled triangle: \cos\theta = \frac{\text{adjacent side}}{\text{hypotenuse}}.	Beninese 4ème focuses on cosine; IB MYP 4 teaches \sin, \cos, and \tan together.
Vector	*Vecteur*	A mathematical object defined by a direction, a sense (orientation), and a magnitude (norm): \vec{u} or \vec{AB}.	Do not confuse *direction* (the line's orientation) with *sens* (arrowhead pointing left/right).
Chasles' relation	*Relation de Chasles*	Fundamental vector addition identity: \vec{AB} + \vec{BC} = \vec{AC}.	Universal in French mathematics; used in IB vector algebra.
Centroid / Centre of gravity	*Centre de gravité (G)*	The concurrency point of the three medians of a triangle, located at 2/3 from each vertex.	In English: centroid. In French: *centre de gravité d'un triangle*.
Expanding brackets	*Développement d'une expression*	Transforming a product into a sum: a(b+c) = ab + ac; (a+b)(c+d) = ac + ad + bc + bd.	In French: *développer et réduire*.
Factoring / Factorisation	*Factorisation*	Transforming a sum into a product of factors: ka + kb = k(a+b).	In French: *factoriser par mise en facteur commun*.

SECTION C — SHARED CONCEPTUAL FOUNDATIONS

Unit Theme: The Pythagorean Metric and Vector Translations

1. Pythagoras' Theorem and its Converse

In any triangle ABC:

- **Direct Theorem**: If triangle ABC is right-angled at A, then:

BC^2 = AB^2 + AC^2

- **Converse Theorem (Réciproque)**: If the lengths of the sides satisfy BC^2 = AB^2 + AC^2, then triangle ABC is right-angled at vertex A.
- **Contrapositive (Conséquence)**: If BC^2 \neq AB^2 + AC^2 where [BC] is the longest side, then triangle ABC is definitely not right-angled.

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A
|\
| \
b | \ c (Hypotenuse)
| \
+----\
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C a B

b^2 + a^2 = c^2

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### 2. Vector Foundations: The Three Defining Attributes

A vector \vec{u} = \vec{AB} is characterised by:

1. **Direction**: The line (AB) or any line parallel to it.
2. **Sense (*Sens*)**: The orientation along the line (from origin A toward terminus B).
3. **Magnitude (*Norme*)**: The geometric length of segment [AB], denoted \|\vec{AB}\| = AB.

Two vectors are equal (\vec{AB} = \vec{CD}) if and only if:

- They share the same direction: (AB) \parallel (CD);
- They share the same sense;
- They have equal lengths: AB = CD.

Geometrically, \vec{AB} = \vec{CD} \iff ABDC is a parallelogram (possibly flattened).

### 3. Worked Example: Method Comparison Across Programmes

**Problem**: A survey team in Dassa measures a triangle with side lengths EF = 65\text{ m}, EG = 52\text{ m}, and FG = 39\text{ m}. Is the survey boundary perpendicular at G?

- **The Beninese APC Deductive Model**:
- *Identifier le plus grand côté* : Le plus grand côté est [EF] avec EF = 65\text{ m}.
- *Calculer séparément* :

EF^2 = 65^2 = 4\ 225

EG^2 + FG^2 = 52^2 + 39^2 = 2\ 704 + 1\ 521 = 4\ 225

- *Constater l'égalité* : On constate que EF^2 = EG^2 + FG^2.
- *Conclusion légale* : D'après la réciproque du théorème de Pythagore, le triangle EFG est rectangle en G. Les limites de parcelles sont donc strictement perpendiculaires en G.
- **The IB MYP Inquiry Model (Criterion A & C)**:
- Test Pythagoras' relation for right triangles: a^2 + b^2 \stackrel{?}{=} c^2.
- Let c = 65, a = 52, b = 39.
- 52^2 + 39^2 = 2\ 704 + 1\ 521 = 4\ 225 = 65^2.
- Since the Pythagorean identity holds, the triangle contains a 90^\circ angle opposite side c. The boundary lines meet at a right angle (90^\circ) at vertex G.

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# SECTION D — CONTEXTUALISED BRIDGING ACTIVITIES

## Activity: Timber Truss Construction in Porto-Novo

### 1. Problem Situation

A master carpenter in Porto-Novo is fabricating a traditional timber roof truss for a community meeting hall.

- The symmetrical king-post truss forms an isosceles triangle ABC with base span BC = 16\text{ metres} and equal rafters AB = AC = 10\text{ metres}.
- The vertical central king-post [AH] joins apex A to the midpoint H of base [BC].
- Two diagonal strut braces [HD] and [HE] are installed, connecting midpoint H to the midpoints D of rafter [AB] and E of rafter [AC].

### 2. The IB Perspective (Inquiry & Trigonometric Modelling)

- In right-angled triangle ABH, midpoint H gives BH = 16 \div 2 = 8\text{ metres}.
- Calculate height AH using Pythagoras:

AH = \sqrt{AB^2 - BH^2} = \sqrt{10^2 - 8^2} = \sqrt{100 - 64} = \sqrt{36} = 6\text{ metres}

- Calculate pitch angle \theta = \widehat{ABH}:

\cos\theta = \frac{BH}{AB} = \frac{8}{10} = 0.8 \implies \theta = \arccos(0.8) \approx 36.87^\circ

- Calculate rafter length using trigonometry to cross-check:
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AB = \frac{BH}{\cos 36.87^\circ} = \frac{8}{0.8} = 10\text{ metres}

3. The Beninese APC Perspective (Concurrence & Midpoint Theorem)

• *Calcul de la hauteur AH* :

Dans le triangle ABH rectangle en H d'après la propriété de la médiatrice :

AB^2 = AH^2 + BH^2 \implies AH^2 = 10^2 - 8^2 = 36 \implies AH = 6\text{ m}

• *Calcul de la longueur du renfort [HD]* :

Dans le triangle ABH rectangle en H, D est le milieu de l'hypoténuse [AB].

Propriété : Dans un triangle rectangle, la longueur de la médiane issue du sommet de l'angle droit est égale à la moitié de la longueur de l'hypoténuse.

HD = \frac{AB}{2} = \frac{10}{2} = 5\text{ mètres}

• *Vérification par le Théorème des milieux* :

Dans le triangle ABC, D est le milieu de [AB] et H est le milieu de [BC]. La droite (DH) est parallèle à (AC) et DH = \frac{AC}{2} = \frac{10}{2} = 5\text{ m}.

4. Comparison of Methods

- IB uses trigonometric cosine ratios to find structural angles.
- Beninese APC leverages the midpoint theorem (*théorème des milieux*) and median property of right triangles, providing an exact geometric proof without decimal rounding.

5. Student Reflection Question

Why does the median of a right-angled triangle equal exactly half of the hypotenuse? How does drawing the circumscribed circle make this obvious?

Espace de réponse:

_____
_____

_____
_____

6. Application in a New Context

A crane boom at the Port of Cotonou has an arm of length 25\text{ m} extending horizontally by 24\text{ m}. Calculate its vertical height above the base pivot and the cosine of the inclination angle.

Working space:

_____
_____

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_____

SECTION E — PROGRESSIVE PRACTICE / EXERCICES PROGRESSIFS

Level 1: Foundational Understanding (Pythagoras & Rational Numbers)

1. In right-angled triangle ABC with hypotenuse [BC], calculate the unknown side:

a) AB = 9\text{ cm}, AC = 12\text{ cm} \implies BC =

b) BC = 25\text{ cm}, AB = 15\text{ cm} \implies AC =

c) AB = 5\text{ cm}, AC = 7\text{ cm} \implies BC = (exact form and 2 decimal places).

2. Calculate and write as an irreducible fraction:

a) A = \frac{2}{3} - \frac{5}{6} \times \frac{4}{7} =

b) B = \left(\frac{1}{2} + \frac{3}{4}\right) \div \left(\frac{5}{8} - \frac{1}{4}\right) =

Level 2: Standard Application (Algebraic Expansion & Exponents)

3. Expand and collect like terms:

a) 3(2x - 5) - 4(x + 2) =

b) (2x + 3)(x - 4) =

c) (3x - 2)(3x + 2) =

4. Simplify using exponent laws:

a) \frac{a^5 \times a^3}{a^2} =

b) (2x^3)^4 =

c) \frac{10^4 \times 10^{-7}}{10^{-2}} =

Level 3: Multi-Step Problems (Equations & Triangle Centres)

5. Solve in \mathbb{Q} (set of rational numbers):

a) 4x - 7 = 2x + 9

b) \frac{2x - 1}{3} = \frac{x + 4}{2}

c) 5(x - 2) - 3(2x + 1) = 0

6. Let ABC be a triangle with medians [AA'], [BB'], and [CC'] intersecting at centroid G.

a) If the total median length AA' = 12\text{ cm}, calculate lengths AG and GA'.

b) State the vector relationship between \vec{GA} and \vec{GA'}.

Level 4: Reasoning & Geometric Proof (Converse Pythagoras & Vectors)

7. Points P, Q, R have coordinates in a plane: PQ = 20\text{ m}, QR = 21\text{ m}, PR = 29\text{ m}.

a) Prove that triangle PQR is right-angled. Name the vertex of the right angle.

b) Calculate \cos\widehat{QPR} and \cos\widehat{PRQ}.

c) Simplify the vector sum: \vec{S} = \vec{PQ} + \vec{QR} + \vec{RP}.

d) Express \vec{QP} + \vec{QR} using a single vector if PQSR is a rectangle.

Level 5: Extension & 4ème National Examination Challenge

8. Situation d'Évaluation Type 4ème (SA N°1 & N°3):

Pour stocker les récoltes de sorgho dans l'Atacora, une coopérative construit un grenier traditionnel modélisé par une pyramide régulière à base carrée SABCD.

- La base carrée ABCD a pour côté AB = 6\text{ mètres}.
- Le sommet S est situé verticalement au-dessus du centre O de la base carrée.
- La hauteur de la pyramide est SO = 4\text{ mètres}.

1. Calcule la longueur de la diagonale [AC] de la base carrée (donne la valeur exacte puis arrondie au centimètre).

2. Calcule la longueur AO.

3. Démontre que le triangle SOA est rectangle en O, et calcule la longueur exacte de l'arête latérale [SA].

4. Soit M le milieu du côté [BC]. Calcule l'apothème SM de la pyramide.

5. Calcule l'aire latérale totale du grenier ainsi que son volume de stockage en \text{m}^3.

SECTION F — INTEGRATED APPLICATION TASKS

Task F-1: Container Transhipment at Cotonou Port (MYP Criterion D & Benin SA 4)

Context: A gantry crane at the Autonomous Port of Cotonou relocates shipping containers from cargo vessels to transport trucks.

- The crane spreader moves a container from point A to point B by translation vector \vec{u}_1.
- A second automated transfer moves the container from B to final storage bay C by vector \vec{u}_2.
- In the port grid coordinate system (units in metres):
- \vec{u}_1 represents a displacement of 30\text{ m} East and 40\text{ m} North.
- \vec{u}_2 represents a displacement of 50\text{ m} East and 20\text{ m} South.

Questions:

1. Calculate the direct distance of displacement from A to B (magnitude \|\vec{u}_1\|) using Pythagoras' Theorem.

2. Express the net resultant displacement vector \vec{AC} using Chasles' relation.

3. Calculate the total horizontal (East) and vertical (North/South) components of vector \vec{AC}.
4. Calculate the straight-line distance between initial position A and final position C.
5. If the container travels directly from A to C at a speed of 1.5\text{ m/s}, calculate the transit time in seconds.

SECTION G — PROGRAMME-SPECIFIC REINFORCEMENT

Part G-1: IB MYP 4 Trigonometry Exploration

In right-angled triangle XYZ right-angled at Y, side XY = 12\text{ cm} and side YZ = 5\text{ cm}.

1. Calculate hypotenuse XZ.
2. State the exact fractional values of:

\sin\hat{X}, \quad \cos\hat{X}, \quad \tan\hat{X}

3. Verify that \tan\hat{X} = \frac{\sin\hat{X}}{\cos\hat{X}}.
4. Verify the Pythagorean identity: \sin^2\hat{X} + \cos^2\hat{X} = 1.

Part G-2: Beninese NPE Vector and Geometry Proof

Exercice de Démonstration Vectorielle (Classe de 4ème)

Soit un parallélogramme ABCD. Soit E le point tel que \vec{CE} = \vec{AB}.

1. Démontre que \vec{AB} = \vec{DC}.
2. Déduis-en que \vec{DC} = \vec{CE}.
3. Que peux-tu en déduire pour le point C par rapport au segment [DE] ? Justifie rigoureusement.

SECTION H — ASSESSMENT, REFLECTION & WORKED SOLUTIONS

Diagnostic Pre-Flight Check

- [ ] I can apply Pythagoras' theorem: c = \sqrt{a^2 + b^2}.
- [ ] I can verify if a triangle is right-angled using the converse theorem.
- [ ] I know how to add vectors using Chasles' relation: \vec{AB} + \vec{BC} = \vec{AC}.
- [ ] I can expand double brackets: (a+b)(c+d).

COMPLETE WORKED SOLUTIONS & MARKING SCHEME

Solutions for Section E (Progressive Practice)

1. **Pythagoras Calculations** (1 mark each):

a) BC = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15\text{ cm}.

b) AC = \sqrt{25^2 - 15^2} = \sqrt{625 - 225} = \sqrt{400} = 20\text{ cm}.

c) BC = \sqrt{5^2 + 7^2} = \sqrt{25 + 49} = \sqrt{74} \approx 8.60\text{ cm}.

2. **Rational Arithmetic** (1 mark each):

a) A = \frac{2}{3} - \frac{20}{42} = \frac{2}{3} - \frac{10}{21} = \frac{14 - 10}{21} = \frac{4}{21}.

b) B = \left(\frac{2+3}{4}\right) \div \left(\frac{5-2}{8}\right) = \frac{5}{4} \div \frac{3}{8} = \frac{5}{4} \times \frac{8}{3} = \frac{10}{3}.

3. **Algebraic Expansions** (1 mark each):

a) 6x - 15 - 4x - 8 = 2x - 23.

b) 2x^2 - 8x + 3x - 12 = 2x^2 - 5x - 12.

c) Difference of squares: 9x^2 - 4.

4. **Exponent Laws** (1 mark each):

a) a^{5+3-2} = a^6.

b) 2^4 x^{3 \times 4} = 16x^{12}.

c) 10^{4 - 7 - (-2)} = 10^{-1} = 0.1.

5. **Equations** (1 mark each):

a) $4x - 2x = 9 + 7 \implies 2x = 16 \implies x = 8$.

b) $2(2x - 1) = 3(x + 4) \implies 4x - 2 = 3x + 12 \implies x = 14$.

c) $5x - 10 - 6x - 3 = 0 \implies -x - 13 = 0 \implies x = -13$.

6. Centroid G (2 marks):

a) In a triangle, the centroid is located at 2/3 of the median from the vertex:

$AG = \frac{2}{3} \times 12 = 8\text{ cm}$, and $GA' = 12 - 8 = 4\text{ cm}$.

b) Opposite senses along the same line: $\vec{GA} = -2\vec{GA'}$.

7. Converse Pythagoras & Vectors (4 marks):

a) Check longest side PR = 29:

$PR^2 = 29^2 = 841$.

$PQ^2 + QR^2 = 20^2 + 21^2 = 400 + 441 = 841$.

Since $PR^2 = PQ^2 + QR^2$, by the converse of Pythagoras' theorem, $\triangle PQR$ is right-angled at Q.

b) $\cos\widehat{QPR} = \frac{PQ}{PR} = \frac{20}{29} \approx 0.690$; $\cos\widehat{PRQ} = \frac{QR}{PR} = \frac{21}{29} \approx 0.724$.

c) By Chasles' relation: $\vec{PQ} + \vec{QR} + \vec{RP} = \vec{PR} + \vec{RP} = \vec{PP} = \vec{0}$ (null vector).

d) In rectangle PQSR, $\vec{QP} + \vec{QR} = \vec{QS}$.

8. Atacora Pyramid Granary (5 marks):

1. In right-angled $\triangle ABC$: $AC^2 = 6^2 + 6^2 = 72 \implies AC = \sqrt{72} = 6\sqrt{2} \approx 8.49\text{ m}$. [1 mark]
2. O is the centre (midpoint of [AC]), so $AO = \frac{AC}{2} = 3\sqrt{2} \approx 4.24\text{ m}$. [1 mark]
3. Height [SO] is perpendicular to base plane: $\triangle SOA$ is right-angled at O.

$SA = \sqrt{SO^2 + AO^2} = \sqrt{4^2 + (3\sqrt{2})^2} = \sqrt{16 + 18} = \sqrt{34} \approx 5.83\text{ m}$. [1 mark]

4. In right-angled $\triangle SOM$ with $OM = \frac{AB}{2} = 3\text{ m}$:

$SM = \sqrt{SO^2 + OM^2} = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5\text{ metres}$. [1 mark]

5. Lateral area: 4 triangles of base 6 and height 5: $\text{Area} = 4 \times \left(\frac{6 \times 5}{2}\right) = 60\text{ m}^2$.

Volume: $V = \frac{1}{3} \times \text{Base Area} \times h = \frac{1}{3} \times (6^2) \times 4 = \frac{1}{3} \times 36 \times 4 = 48\text{ m}^3$. [1 mark]

Solutions for Section F (Task F-1: Cotonou Port Spreader)

1. Magnitude $|\vec{u}_1| = \sqrt{30^2 + 40^2} = \sqrt{900 + 1\ 600} = \sqrt{2\ 500} = 50\text{ metres}$. [2 marks]
2. Chasles' relation: $\vec{AC} = \vec{AB} + \vec{BC} = \vec{u}_1 + \vec{u}_2$. [1 mark]
3. Total horizontal: $30 + 50 = 80\text{ m}$ East. Total vertical: $40 - 20 = 20\text{ m}$ North. [2 marks]
4. Straight-line distance: $AC = \sqrt{80^2 + 20^2} = \sqrt{6\ 400 + 400} = \sqrt{6\ 800} \approx 82.46\text{ metres}$. [2 marks]
5. Transit time: $t = \frac{82.46}{1.5} \approx 54.97\text{ seconds}$ ($\approx 55\text{ s}$). [1 mark]

Solutions for Section G-2 (Benin Vector Proof 4ème)

1. ABCD est un parallélogramme, donc par définition de l'égalité vectorielle, les côtés opposés orientés sont égaux : $\vec{AB} = \vec{DC}$. [1 pt]
2. Par hypothèse, $\vec{CE} = \vec{AB}$. Comme $\vec{AB} = \vec{DC}$, on en déduit par transitivité que $\vec{DC} = \vec{CE}$. [2 pts]
3. L'égalité vectorielle $\vec{DC} = \vec{CE}$ signifie que les vecteurs ont la même direction (les droites (DC) et (CE) sont confondues, donc les points D, C, E sont alignés), le même sens, et la même longueur (DC = CE). Par conséquent, le point C est le milieu du segment [DE]. [2 pts]

End of Workbook 4 (Quatrième / MYP 4).